

Microbiome Analysis Guide

Diversity Metrics

Alpha Diversity

Purpose: Measures of alpha diversity give us an understanding of the diversity within a sample (intra-sample diversity). All of the commonly used metrics take into account species richness and some metrics also into account the relative abundance of different species within a given sample.

Types of Alpha Diversity Metrics: [Shannon](#), [Simpson's](#), [Chao-1](#)

Typical Visualizations: [Boxplots](#), [Density Plots](#), [Scatterplots](#), [Violin Plots](#)

R packages: [vegan](#), [ggplot2](#), [alpha.div](#)

Beta Diversity

Purpose: Beta diversity metrics give us information about the diversity between samples (inter-sample diversity). Using these measures, we can determine the level of similarity between all samples in a dataset.

Types of Beta Diversity Metrics: [Jaccard Distance](#), [Bray-Curtis Dissimilarity](#), [Weighted/Unweighted Unifrac](#)

Typical Visualization: [tSNE](#), [PCA](#), [UMAP](#), [NMDS](#)

R packages: [lme4](#), [Bioconductor: scater](#)

Differential Abundance Analysis

Purpose: Differential abundance metrics can be used to identify specific organisms enriched or depleted in a subset of samples. For example, these metrics can be used to identify organisms enriched in a disease state, or positively correlated with a certain treatment.

Typical Visualizations: Tabular, [volcano plot](#), [MA-plot](#)

R packages: [DESeq2](#), [limma-voom](#), [ANCOM-BC](#), [edgeR](#)

Correlation Analysis

Purpose: Correlation analysis can be used to determine how often organisms co-occur. This information can be used to identify symbiotic or antagonistic relationships between species or groups of organisms..

Typical Visualizations: [Pearson/Spearman Correlations](#), [Correlation Matrices](#)

R packages: [ggpubr](#), [ggcorrplot](#)

Network Analysis

Purpose: Network analysis is often used to visualize the relationships between microbial species, both positive and negative, at a community level. This can be used to find important interactions between microbes in health and disease.

Typical Visualizations: [Networks](#)

R packages: [igraph](#), [gggraph](#), [tidygraph](#), [network](#)

MicrobiomeDB

Purpose: A resource generated by our collaborators at the University of Pennsylvania, MicrobiomeDB is a one-stop-shop for microbiome analysis. It accepts a .biom file as input and can be used to build relative abundance plots, and perform diversity analysis (alpha/beta), differential abundance analysis and correlation analysis.

Note: BIOM files can be downloaded directly from CZID. Select your samples of interest and click the download icon. Scroll down and select the “Combined Microbiome File” option. Select your Download Metric from the dropdown menu. We suggest using NT RPM. You can also add filters to limit the output to hits that meet specific criteria. We suggest using the following three filters: NT RPM ≥ 10 , NR RPM ≥ 1 , NT L ≥ 50 . This will only show taxa with at least 10 reads per million at the nucleotide level, at least one read per million at the protein level and with a minimum alignment length of 50 bases. This filtering prevents the reporting of spurious taxa hits.

<https://microbiomedb.org/mbio/app>